

Study the properties of the following systems!

Example 1.

$$y[n] = \frac{1}{2}(x[2n] + x[2n + 1]).$$

- **Causality.** For $n > 0$, the output $y[n]$ depends on $x[2n]$ and $x[2n + 1]$ with $2n > n$ and $2n + 1 > n$, hence on future input samples. $\Rightarrow$ **not causal**
- **Memorylessness.** A memoryless system satisfies: if $x_1[n] = x_2[n]$ then $y_1[n] = y_2[n]$ for the same n . Here $y[n]$ depends on $x[2n]$ and $x[2n + 1]$, i.e. on indices different from n in general. $\Rightarrow$ **not memoryless**
- **Linearity.**

– **Homogeneity.** For scalar a ,

$$\mathcal{H}(ax)[n] = \frac{1}{2}(ax[2n] + ax[2n + 1]) = a \mathcal{H}(x)[n].$$

– **Additivity.** For x_1, x_2 ,

$$\mathcal{H}(x_1 + x_2)[n] = \frac{1}{2}(x_1[2n] + x_2[2n] + x_1[2n + 1] + x_2[2n + 1]) = \mathcal{H}(x_1)[n] + \mathcal{H}(x_2)[n].$$

$\Rightarrow$ **linear**

- **Time invariance.** Let $x_s[n] = x[n - d]$. Then

$$\mathcal{H}(x_s)[n] = \frac{1}{2}(x[2n - d] + x[2n + 1 - d]),$$

while the shifted output is

$$y[n - d] = \frac{1}{2}(x[2(n - d)] + x[2(n - d) + 1]) = \frac{1}{2}(x[2n - 2d] + x[2n - 2d + 1]).$$

These differ in general, so $\Rightarrow$ **time-varying (not time-invariant)**.

- **BIBO stability.** If $|x[k]| \leq B$ for all k , then

$$|y[n]| \leq \frac{1}{2}(|x[2n]| + |x[2n + 1]|) \leq \frac{1}{2}(B + B) = B.$$

$\Rightarrow$ **BIBO stable**

Example 2.

$$y[n] = \frac{3x[n-1] + x[n-2]}{x[n-3]}.$$

- **Causality.** The output depends only on $x[n-1], x[n-2], x[n-3]$ (past values).
 $\Rightarrow$ **causal (where defined)**
- **Memorylessness.** The output uses delayed samples $(n-1, n-2, n-3)$, hence depends on values other than $x[n]$. $\Rightarrow$ **not memoryless**
- **Linearity.**

– **Homogeneity.** For scalar a ,

$$\mathcal{H}(ax)[n] = \frac{3ax[n-1] + ax[n-2]}{ax[n-3]} = \frac{3x[n-1] + x[n-2]}{x[n-3]} = \mathcal{H}(x)[n] \neq a\mathcal{H}(x)[n]$$

in general. $\Rightarrow$ **fails homogeneity**

– **Additivity.** The input appears in the denominator, so in general

$$\mathcal{H}(x_1 + x_2)[n] \neq \mathcal{H}(x_1)[n] + \mathcal{H}(x_2)[n].$$

$\Rightarrow$ **fails additivity**

$\Rightarrow$ **nonlinear**

- **Time invariance.** Let $x_s[n] = x[n-d]$. Then

$$\mathcal{H}(x_s)[n] = \frac{3x[n-1-d] + x[n-2-d]}{x[n-3-d]}.$$

With the substitution $m = n - d$, this becomes

$$\mathcal{H}(x_s)[n] = \frac{3x[m-1] + x[m-2]}{x[m-3]} = y[m] = y[n-d].$$

$\Rightarrow$ **time-invariant (where defined)**

- **BIBO stability.** Take a bounded input with $|x[k]| \leq 1$ for all k , but choose some index k_0 with $x[k_0] = \varepsilon$ ($0 < \varepsilon \ll 1$) and $x[k] = 1$ otherwise. Then

$$y[k_0 + 3] = \frac{3x[k_0 + 2] + x[k_0 + 1]}{x[k_0]} = \frac{4}{\varepsilon} \rightarrow \infty \quad (\varepsilon \rightarrow 0),$$

hence $\Rightarrow$ **not BIBO stable.**

Example 3.

$$y[n] = x[n] e^{n/14}.$$

- **Causality.** The output depends only on $x[n]$. $\Rightarrow$ **causal**
- **Memorylessness.** The output at time n depends only on the input at the same time n . $\Rightarrow$ **memoryless**
- **Linearity.**

– **Homogeneity.** For scalar a ,

$$\mathcal{H}(ax)[n] = ax[n]e^{n/14} = a \mathcal{H}(x)[n].$$

– **Additivity.** For x_1, x_2 ,

$$\mathcal{H}(x_1 + x_2)[n] = (x_1[n] + x_2[n])e^{n/14} = \mathcal{H}(x_1)[n] + \mathcal{H}(x_2)[n].$$

$\Rightarrow$ **linear**

- **Time invariance.** Let $x_s[n] = x[n - d]$. Then

$$\mathcal{H}(x_s)[n] = x[n - d]e^{n/14}, \quad y[n - d] = x[n - d]e^{(n-d)/14}.$$

Since $e^{n/14} \neq e^{(n-d)/14}$ in general, $\Rightarrow$ **time-varying (not time-invariant)**.

- **BIBO stability.** For bounded input $|x[n]| \leq B$,

$$|y[n]| = |x[n]|e^{n/14} \leq Be^{n/14},$$

which is unbounded as $n \rightarrow \infty$. $\Rightarrow$ **not BIBO stable**